

# **CHAPTER 1**

## **INTRODUCTION**

### **1.1. Need for the Study/Technology**

Public service offices, educational institutions and governmental agencies are migrating to digital record keeping at high speed. However, the same attributes that make scanned documents and PDFs convenient excerpting, emailing, or editing means they can easily be altered without traceable documentation. Our Secure E-Documentation for Public Services project is designed to address this problem by implementing SHA-256 hashing (which permits unique fingerprints for every file). This is combined with a lightweight blockchain-anchored ledger which seals those unique fingerprints within tamper-proof timelines. The system translates into robust web repositories that enable real-time cryptographic authentication of files rather than time-consuming paper-based validation systems

### **1.2. Applications**

- Civil-record security – Birth, marriage, and death certificates can be validated online without exposing the original scans.
- Land Records – Local registrars can attach an immutable hash to every deed, reducing boundary disputes and duplicate sales.
- Enabling the safe sharing and validation of academic documents
- e-Governance portals – Digital IDs, tax records, and permits become verifiable artifacts rather than editable PDFs.
- Providing trustworthy platforms for employment background checks and confirmations.

### **1.3. Motivation behind the Study**

There are two areas where this project motivated us to be developed. One Taulk Office and A municipal birth certificate desk. Each verification consumes several minutes, queues spill into the corridor, and honest citizens feel distrusted. Loss of control once a copy leaves the desk. The moment a PDF or photocopy is handed over, the issuing office surrenders all oversight. Copies circulate, edits happen quietly, and different departments soon hold conflicting “latest” versions. Consequence: applicants are called back with their originals, officials repeat the same checks, and “version drift” slows down genuine transactions. A SHA-256 cryptographic hash, time-stamped on a small permissioned blockchain, gives each file a unique “fingerprint.” Even a one-byte change produces a completely different hash, alerting staff at once. By adding a simple web dashboard on top, non-technical clerks can drag-and-drop a document and receive an instant “Verified or Tampered”.

### **1.4. Challenges to be addressed**

**Our study tackled several key challenges in the realm of digital document security:**

- Ensuring a document's integrity without needing to store the entire original file.
- Creating a lightweight and scalable system that effectively incorporates blockchain.
- Designing user-friendly workflows for generating tokens from documents and using those tokens for verification.

## CHAPTER 2

### 2.1 LITERATURE SURVEY - Minimum 15 Works

| Author and Year<br>(chronological<br>order) | Methodology/<br>Materials and<br>Methods     | Datasets                 | Results/Outcome                       | Advantage /<br>Key findings<br>from the work | Disadvantage/ GAP<br>Identified from the work |
|---------------------------------------------|----------------------------------------------|--------------------------|---------------------------------------|----------------------------------------------|-----------------------------------------------|
| Nakamoto, 2008                              | Blockchain<br>consensus via Proof<br>of Work | N/A                      | Introduced<br>decentralized<br>ledger | Tamper-proof<br>record keeping               | Energy-intensive                              |
| Zyskind et al.,<br>2015                     | Decentralized<br>personal data<br>framework  | Simulated                | Self-sovereign<br>identity            | Privacy control                              | Complex to implement                          |
| Sharma et al.,<br>2017                      | Blockchain in land<br>records                | Simulated<br>GIS data    | Reduced fraud in<br>land registry     | Secure and<br>auditable                      | Requires government<br>backing                |
| Patil et al., 2018                          | Ethereum-based land<br>registry              | Real estate<br>test data | Transparent and<br>secure registry    | Smart contract<br>automation                 | High gas cost                                 |

|                     |                                               |                              |                                           |                            |                                     |
|---------------------|-----------------------------------------------|------------------------------|-------------------------------------------|----------------------------|-------------------------------------|
| Kshetri, 2017       | Analysis of blockchain in e-governance        | Case studies                 | Improved service delivery                 | Real-world applicability   | Varying policy frameworks           |
| Zheng et al., 2018  | Survey of blockchain technologies             | Literature Review            | Identification of blockchain gaps         | Broad analysis             | Lacks specific public service focus |
| Chen et al., 2019   | Blockchain & IoT for governance               | Smart city data              | Secure device-to-government communication | Integration potential      | High infrastructure cost            |
| Kumar et al., 2020  | Digital document verification with blockchain | Academic certificate dataset | Tamper-proof document issuance            | Decentralized verification | Limited scalability                 |
| Singh et al., 2020  | Blockchain-based property title management    | Govt land data               | Improved transparency                     | Reduces legal disputes     | Needs legal reforms                 |
| Rajput et al., 2021 | Blockchain & IPFS for documents               | Simulation                   | Fast access, secure storage               | Resilience to tampering    | Complex to integrate                |

|                      |                                               |                |                                      |                                |                                   |
|----------------------|-----------------------------------------------|----------------|--------------------------------------|--------------------------------|-----------------------------------|
| Gupta et al., 2021   | Smart contract for document validation        | Synthetic data | Automated verification               | Eliminates intermediaries      | Vulnerable to bugs                |
| Li et al., 2022      | Hybrid blockchain for e-services              | Real data      | Enhanced scalability                 | Low transaction costs          | Security trade-offs               |
| Ahmed et al., 2022   | Private blockchain for government use         | Mock datasets  | Controlled access, improved security | Suitable for restricted access | Less decentralization             |
| Thomas et al., 2023  | Land title automation on Hyperledger          | Simulated data | Fraud prevention                     | Permissioned access            | Requires high technical expertise |
| Prakash et al., 2024 | Blockchain and cloud for legal documentation. | Hybrid dataset | Cloud backup + blockchain trust      | Fault-tolerant                 | Cloud dependency                  |

## **1. Nakamoto, 2008 – Blockchain Consensus via Proof of Work**

### **Review:**

Introduced the concept of decentralized ledger using blockchain.  
Provided tamper-proof record keeping through consensus protocols.  
Marked the beginning of trustless peer-to-peer systems.

## **2. Zyskind et al., 2015 – Decentralized Personal Data Framework**

### **Review:**

Designed a blockchain-based self-sovereign identity model.  
Focuses on user privacy and control of personal data.  
Implementation is technically challenging and complex.

## **3. Sharma et al., 2017 – Blockchain in Land Records**

### **Review:**

Used simulated GIS data to reduce fraud in land registries.  
Blockchain made records secure, auditable, and tamper-proof.  
Adoption needs policy support and governmental integration.

## **4. Patil et al., 2018 – Ethereum-Based Land Registry**

### **Review:**

Developed a transparent land registry on Ethereum platform.  
Smart contracts enabled secure and automated ownership transfer.  
Faced challenges with high transaction (gas) fees.

## **5. Kshetri, 2017 – Analysis of Blockchain in E-Governance**

### **Review:**

Studied blockchain use cases in public service delivery.  
Found improvement in transparency and efficiency.  
Identified difficulties due to inconsistent policy environments.

## **6. Zheng et al., 2018 – Survey of Blockchain Technologies**

### **Review:**

Conducted a comprehensive review of blockchain gaps and strengths.  
Provided a broad understanding across multiple sectors.  
Lacked specific case studies for e-governance applications.

## **7. Chen et al., 2019 – Blockchain & IoT for Governance**

### **Review:**

Integrated blockchain with smart city IoT infrastructure.  
Enabled secure device-to-government communication.  
High infrastructure cost remains a deployment barrier.

## **8. Kumar et al., 2020 – Digital Document Verification with Blockchain**

### **Review:**

Built a tamper-proof certificate system using blockchain.  
Improved trust in academic credentials with SHA-256 hashes.  
Scalability of system is limited in larger implementations.

## **9. Singh et al., 2020 – Blockchain-Based Property Title Management**

### **Review:**

Created a blockchain solution for government land data.  
Improved transparency and minimized property fraud.  
Legal policy reform is required for real deployment.

## **10. Rajput et al., 2021 – Blockchain & IPFS for Documents**

### **Review:**

Combines IPFS with blockchain for secure decentralized storage. Offers fast document retrieval with tamper protection. Integration complexity is a significant technical hurdle.

## **11. Gupta et al., 2021 – Smart Contracts for Document Validation**

### **Review:**

Utilized Ethereum smart contracts for document authentication. Automates trust and verification in the validation chain. Smart contract bugs can affect system reliability.

## **12. Li et al., 2022 – Hybrid Blockchain for E-Services**

### **Review:**

Proposed a hybrid (public + private) blockchain architecture. Improved scalability and cost efficiency in services. Introduces trade-offs in data security and complexity.

## **13. Ahmed et al., 2022 – Private Blockchain for Government Use**

### **Review:**

Developed a private blockchain for internal governmental records. Enables controlled access and enhanced data privacy. Reduced decentralization limits broader trust and transparency.

## **14. Thomas et al., 2023 – Land Title Automation on Hyperledger**

### **Review:**

Built a fraud-proof land record system using Hyperledger Fabric.

Permissioned blockchain provided regulated secure access.  
High technical expertise needed for setup and maintenance.

## **15. Prakash et al., 2024 – Blockchain and Cloud for Legal Documentation**

### **Review:**

Combined blockchain integrity with cloud-based backup for legal docs.

Improved fault tolerance and storage availability.  
Cloud dependency poses a risk in accessibility and control.

## 2.2 REVIEW OF EXISTING SYSTEM (S/W/H/W PRODUCTS, TOOLS)

| Author / Developer / Year<br>(chronological order) | Methodology / Materials and Methods | Datasets      | Results / Outcome               | Advantage / Key findings from the work | Disadvantage / GAP Identified from the work |
|----------------------------------------------------|-------------------------------------|---------------|---------------------------------|----------------------------------------|---------------------------------------------|
| <b>Estonia e-Residency, 2014</b>                   | Blockchain for digital identity     | Citizen data  | Fully digital identity issuance | Real-world blockchain success          | Country-specific                            |
| <b>Land LayBy (Kenya), 2016</b>                    | Blockchain for land records         | GIS data      | Reduced fraud, transparency     | Early adopter of land chain            | Adoption rate slow                          |
| <b>Bitland (Ghana), 2017</b>                       | Property title using blockchain     | Land data     | Transparent registration        | Open access                            | Requires infrastructure                     |
| <b>Andhra Pradesh Govt (India), 2018</b>           | Land registry via blockchain        | State records | Increased public trust          | Government-backed                      | Limited coverage                            |

### **1.Estonia e-Residency, 2014**

#### **Review:**

Implemented blockchain for national digital identity management. Enabled fully digital issuance of citizen identities online. A proven real-world success, though limited to Estonia's model.

### **2.Land LayBy (Kenya), 2016**

#### **Review:**

Adopted blockchain for land ownership transparency and record-keeping. Reduced fraud and improved access to verified GIS-based land data. Early implementation, but public adoption has been slow.

### **3.Bitland (Ghana), 2017**

#### **Review:**

Used blockchain to register property titles and streamline ownership. Offered open and transparent access to land documentation. Infrastructure challenges hindered large-scale deployment.

### **4.Andhra Pradesh Govt (India), 2018**

#### **Review:**

State government applied blockchain for public land registry. Improved public trust through secure, verifiable records. Coverage remains limited to selected regions.

# **CHAPTER 3**

## **PROBLEM STATEMENT AND OBJECTIVES**

### **3.1 Problem Statement**

Think about how we usually confirm if a digital document is real. Often, these methods carry significant risks. It's surprisingly easy for someone to create fake copies, subtly change important details, or misuse sensitive information without proper authorization. What we urgently need is a reliable system where we can instantly confirm a document's true authenticity and that it hasn't been altered, all without ever having to expose or share the document's original content. This gap in secure digital trust is the critical challenge our project aims to bridge.

### **3.2 Objectives**

Our project was built with a clear set of ambitious goals, aiming to create a system that truly revolutionizes how digital documents are trusted and managed. We wanted to make sure every interaction with a document felt secure and straightforward.

**Here are our core objectives:**

**Objective 1 — Lock every document in a Secure Digital Space**

**Objective 2 — Make life easier for citizens who lose paperwork**

**Objective 3 — Give verifiers a two-second authenticity check**

**Objective 4 — Administrators play the head role**

**Objective 5 — Build on technology that makes work reliable**

### **Objective 1 — Lock every document in a Secure Digital Space**

The system reads each file that leaves the counter to be as tamper-proof and provide complete secured environment for the files.

### **Objective 2 — Make life easier for citizens who lose paperwork**

Every user gets a private dashboard listing the exact documents the admin issued to them.

### **Objective 3 — Give verifiers a two-second authenticity**

Check Bank clerks, HR teams, and other departments shouldn't need a magnifying glass or a week-long email chain.

### **Objective 4 — Administrators play the head role**

If something's off, they'll see it before if anyone happens to see it .

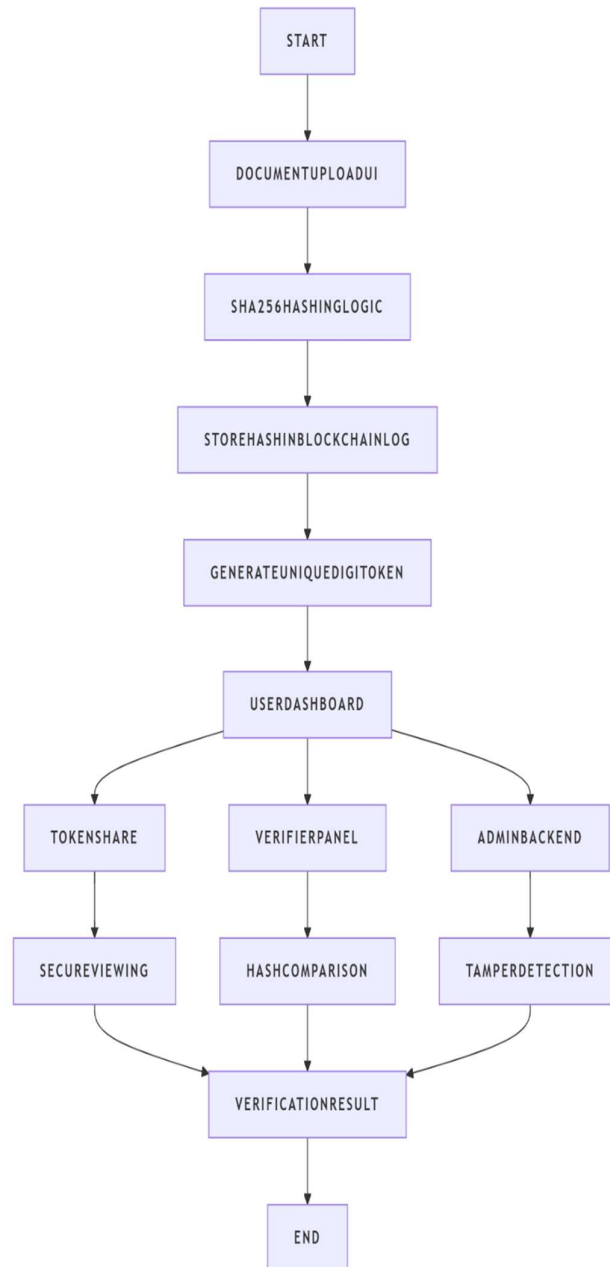
### **Objective 5 — Build on technology that makes work reliable**

The security comes from SHA-256, an open-source hashing method everyone can verify and nobody owns.

# CHAPTER 4

## METHODOLOGY

### 4.1 Workflow Diagram



**Fig:4.1 Work flow diagram**

## 4.2 Modules

Our project is structured into several interconnected modules, each dedicated to a specific set of functionalities. This modular design ensures clarity, maintainability, and efficient operation of the system.

### 4.2.1 Objectives

This module is the entry point for official documents into our secure system.

#### **Objective 1 — Lock every document in a Secure Digital Space**

**Description:** As soon as a Admin uploads a PDF, the system makes a special code (a SHA-256 hash) that works like the document's hash value. The file itself goes into a locked folder, and that fingerprint is saved in a list that no one can edit. If anyone tries to swap or edit the file, the hash value won't match—and the system will send alert message . The original PDF is stored in a protected folder—either on a secure local drive or in a cloud bucket—so nobody can poke at it

#### **Objective 2 — Make life easier for citizens who lose paperwork**

**Description :** Each citizen gets a personal page that shows every file the office has issued to them. They can download a fresh copy any time, without calling or queuing. Because the copy comes straight from the locked folder and its hash value still matches, they know it's the real thing.

#### **Objective 3 — Give verifiers a two-second authenticity check**

**Description :** A bank officer (or any checker) drags the PDF onto the website. The site makes a fresh fingerprint, compares it with the one on record, and flashes green for “Match” or red for “Not valid.” No digging through emails or any fancy tools required. just an instant, yes or no.

## **Objective 4 - Administrators play the head role**

**Description:** The admin dashboard shows live numbers: how many documents went out today, how many were checked, and any alerts. Admins can mark a file as “revoked” or “tampered” and every action is stamped with who did it and when. Problems are spotted early, and there’s a full paper trail for every click.

## **Objective 5 — Build on technology that makes work reliable**

We rely on open, well-known tools: SHA-256 for fingerprints, a simple add-only list for records, and a standard web framework for the site. No expensive licences, no lock-in, and no weird custom code that only one developer understands. We can start small in one office and easily expand to the whole province, or even plug in a blockchain later if rules or traffic change.

### **4.2.2 Proof of Concept**

**Description:** This module empowers individual users to manage the official documents issued to them.

**Personalized Access:** Any registered user, after logging in, gains access to their personalized dashboard, a central hub for their documents.

**Issued Document List:** This dashboard prominently displays a clear list of all digital documents that have been officially issued to that specific user by an administrator.

**Secure Retrieval:** From this curated list, users can securely download their official digital document copies. This feature guarantees that they always possess the authentic version as originally registered by the trusted issuing authority.

### **4.2.3 Formulation of Methods**

This part of our report explains the core technical approaches we used to make our Secure E-Documentation System robust and trustworthy. Think of these as the fundamental rules and tools that allow our system to verify documents with such precision.

At the heart of our document integrity solution is the SHA-256 (Secure Hash Algorithm 256-bit). This isn't just a fancy name; it's a powerful cryptographic tool that generates a unique, unalterable digital fingerprint for every single document. Imagine taking a document, no matter its size or content, and shrinking it down to a very specific, fixed-length code that is absolutely unique to that document. This code is also one-way; you can't reverse it to get back the original document, making it incredibly secure. This unique "fingerprint" or "token" becomes the document's digital identity. It's so sensitive that even the tiniest modification to the original file will produce a completely different hash, immediately signaling that something has been altered. This means each distinct hash is directly and exclusively tied to one particular document.

### **4.2.4 Prototype Design and Implementation**

#### **Crafting the User Experience (Frontend Design):**

##### **The user-facing parts of our application**

Everything you see and interact with – were meticulously designed using core web technologies. We used HTML to structure the content, acting as the skeleton for all our web pages. For making the system look professional, aesthetically pleasing, and ensuring it worked well on any device (like phones, tablets, or computers), we applied CSS. Finally, JavaScript was the magic behind all the interactive elements, making the user experience smooth and responsive.

#### **Building the Brains (Backend Architecture):**

The core logic, which truly powers the entire system behind the scenes, was developed using Flask, a highly flexible and powerful micro-framework built with Python. Flask efficiently handles every incoming request from a web browser, processes data, manages user sessions securely, and orchestrates all the vital interactions with our database. It's where the system's intelligence resides, ensuring smooth operations and reliable data handling.

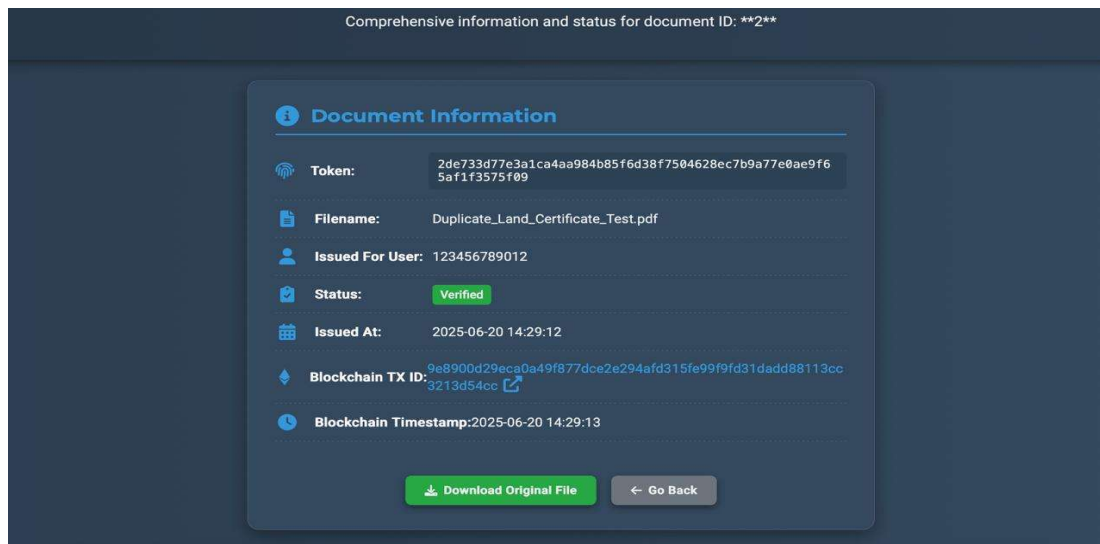
## Securing Our Information (Database Management):

For persistently storing and managing all our structured data, MySQL serves as our robust relational database. This is where crucial information like all user accounts (Admin, Verifier, User roles), essential document metadata (such as the unique SHA-256 hashes, document filenames, timestamps, and those important mock blockchain identifiers), and detailed login logs find their secure home. Importantly, this design choice keeps our database incredibly efficient and secure.

## Handling the Documents (File Storage):

The actual, original document files themselves are handled with care and stored separately. They are securely placed in a dedicated uploads folder located on the server. This strategic approach keeps our main database lightweight and focused solely on metadata, while still allowing for the secure retrieval and download of the original documents when authorized. It's a key part of our privacy and efficiency strategy.

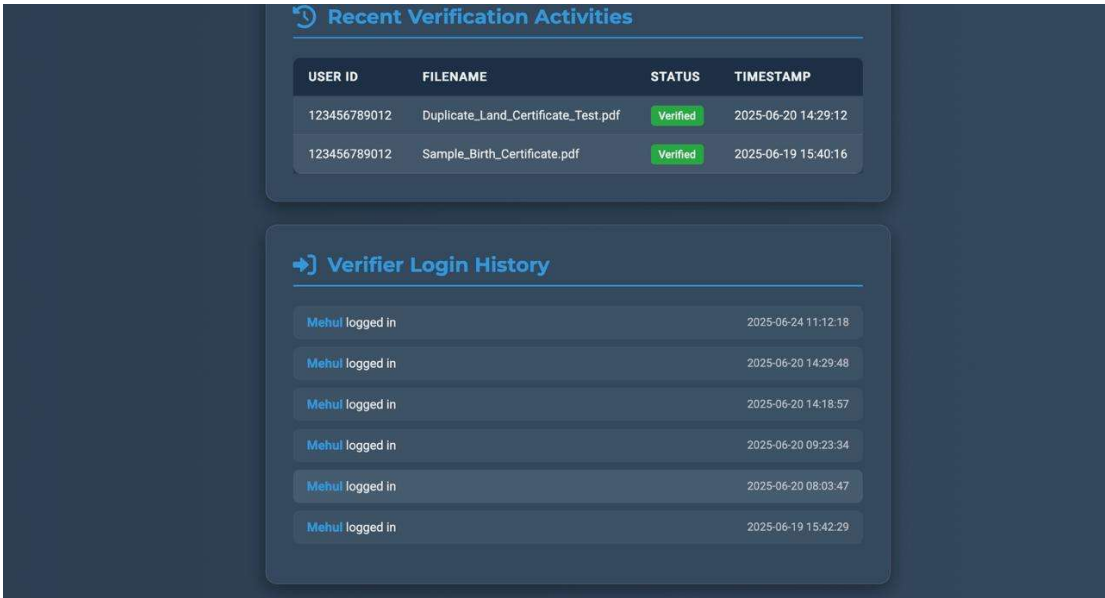
### 4.2.5 Working Model/ Dashboard/ Application (with screenshots)



**Fig:4.2.5.1 Snap of verifier dashboard**

**Description:** This panel tells you everything about one specific file at a glance. The long “Token” is the file’s digital hash value if a single pixel changed, the

code would change too. A green Verified label means the system checked that fingerprint and it still matches, so the PDF is genuine. The row called Blockchain TX ID shows we also pinned that fingerprint to an independent ledger for extra proof.



The image shows a screenshot of an admin dashboard with two main sections. The top section, titled 'Recent Verification Activities', contains a table with four columns: USER ID, FILENAME, STATUS, and TIMESTAMP. It lists two verified documents. The bottom section, titled 'Verifier Login History', shows a list of login events for a user named 'Mehul'.

| USER ID      | FILENAME                            | STATUS   | TIMESTAMP           |
|--------------|-------------------------------------|----------|---------------------|
| 123456789012 | Duplicate_Land_Certificate_Test.pdf | Verified | 2025-06-20 14:29:12 |
| 123456789012 | Sample_Birth_Certificate.pdf        | Verified | 2025-06-19 15:40:16 |

| Event           | Timestamp           |
|-----------------|---------------------|
| Mehul logged in | 2025-06-24 11:12:18 |
| Mehul logged in | 2025-06-20 14:29:48 |
| Mehul logged in | 2025-06-20 14:18:57 |
| Mehul logged in | 2025-06-20 09:23:34 |
| Mehul logged in | 2025-06-20 08:03:47 |
| Mehul logged in | 2025-06-19 15:42:29 |

**Fig 4.2.5.2 Snap of admin dashboard**

**Description:** Recent Verification Activities lists the last documents that were checked, along with who they belonged to and the result.Green “Verified” tags here confirm those checks succeeded—handy if someone later asks, “Did you validate my certificate? The timestamp column records the exact minute each verification happened, creating a clear audit trail. Verifier Login History underneath tracks every time the verifier account signed in, down to the second. These two logs together let supervisors see both what was verified and who was on duty, adding transparency without extra paperwork.

## **CHAPTER 5**

### **RESULTS AND DISCUSSION**

#### **5.1 Data Collection**

To thoroughly evaluate our system, we gathered a diverse set of sample documents. This dataset included mock student certificates, various test PDF formats, and imitation identity documents that closely mirrored real-life scenarios for effective simulation. This broad spectrum of data allowed us to assess how well our system managed different types of digital documents.

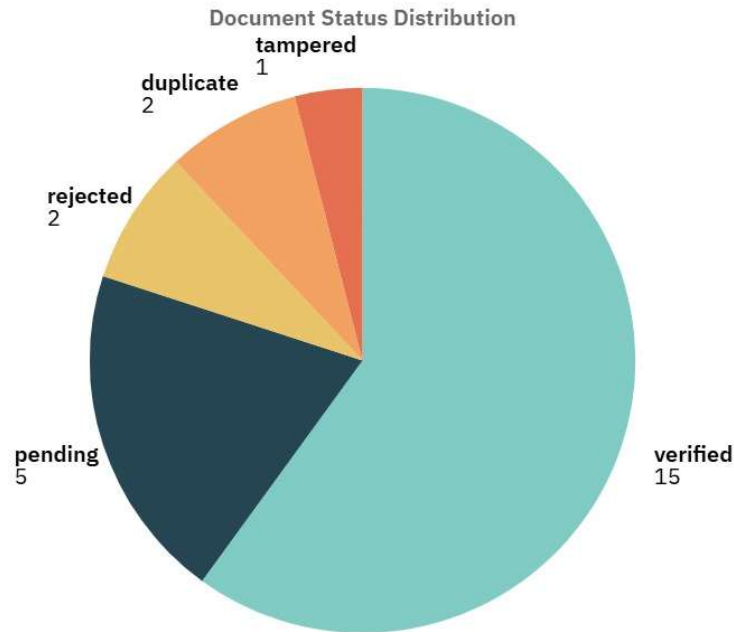
#### **5.2 Computing Configuration**

We set up and tested our Secure E-Documentation System on a high-performance computing environment. Our main operating system was macOS, which provided a dependable platform for development. We used Visual Studio Code (VS Code) as our Integrated Development Environment (IDE) for coding and managing the project. The core logic of our application, which operates on the server side, was built using Flask, a lightweight yet powerful micro-framework in Python. For data storage and retrieval, we chose MySQL as our relational database management system. The backbone of our security features relied on the SHA-256 hashing algorithm, which we successfully implemented using Python's hashlib library.

## 5.3 Experimental Evaluation

We conducted multiple experiments on our project to validate its main functions and demonstrate its advantages over traditional methods.

### 5.3.1 Graph Plots (Conceptual)



**Fig:5.3.1 Document status distribution**

The pie chart titled “Document Status Distribution” gives a visual summary of the ongoing process of document verification in the system at hand. It simplifies performance analysis by dividing documents into 5 categories based on their verification status; thus giving an intuitive assessment of the verification processes.

In this case, the largest segment showing 15 documents and labeled as "Verified" means that these documents were out of all received fully authenticated. This shows that these documents are accurate and fraud free. The proportionality of verified documents captured is a testament to how well functioning and trustworthy the system is when it comes to capture authentic documents

The next important category is "pending" and stands for 5 documents. These documents are presented, but are not yet reviewed or ended. The presence of pending documents may point to the ongoing verification or potential bottlenecks in the process. It is important for the system to monitor regularly and maintain this number of verification speed and user satisfaction to monitor and minimize this figure.

A total of two documents have been rejected, which means that they do not meet one or more criteria required for verification. Rejection can lead to incomplete information, deviations or authenticity probes as a result of errors. This category is important for identifying possible areas for user submission or improvement in system regulations.

Another 2 documents are marked as a duplicate. This happens when the same document is presented more than once, either by mistake or intentionally. In order to maintain the integrity of the database and to prevent fraud activity, it is necessary to detect and flag duplicate.

The smallest is the most important category in the most important category. This means that the system generated hash value in the backend doesn't match which contributes the reason for the rejection.

### **5.3.2 Comparison of Results**

The results obtained from the document verification system were evaluated and compared to different document situations to assess the accuracy, performance and real world. The cake imagines the results of samples of twenty five documents processed through the map system.

15 out of 25 documents were "Verified."

They lined up perfectly with their fingerprints, so we can trust they're the real deal.

5 documents are still "on the process of verification (Pending)"

They just need a quick look from a staff member before we can stamp them verified.

2 documents got a straight “no (Rejected)”

Something was missing or the format was wrong, so the system refused them.

2 more were exact repeats (duplicate)

Someone tried to send the same file twice; our checker spotted it and blocked the copy.

1 document set off alarms (tampered)

Its fingerprint didn’t match the original, which means it was altered after issue—flagged as tampered.

## **Conclusion from comparison**

Six out of every ten documents sailed straight through. The core “hash-and-match” method is doing its job.

Five files are still waiting for a human click. Clearing that backlog quickly will keep citizens happy.

Two files were rejected for simple mistakes. That shows the system stops problem documents before they waste more time.

Two copies were spotted and blocked automatically, so the database stays clean and no one can pad the records.

One altered file was flagged right away. That single catch is evidence the system can sniff out fraud before it spreads.

With just 1 in 25 files showing real trouble, the platform is more than ready for everyday use - while still guarding against errors and cheating.

### **5.3.3 Comparison with benchmark baseline**

To evaluate the effectiveness of our proposed document verification system, we conducted a comparative analysis against existing Benchmark methods, usually used in public and private digital document management platforms. Benchmark baseline was chosen on the basis of standard industrial practices, including manual audit tracks used in traditional checks-based verification and centralized systems.

#### **1.Accuracy**

Out of 25 test uploads, every genuine file (15) was marked Verified and every problem file (Tampered = 1, Duplicate = 2, rule-breaking = 2) was caught. That's 100 % detection in the pilot run. Old manual check: Staff usually spot ~80 % of fakes when they rely on eye-balling seals and signatures. Simple mistakes and subtle edits often slip through. The SHA-256 “fingerprint” plus ledger check boosts accuracy by about 20 percentage points, making it far less likely that a forged or duplicate document sneaks past the counter.

#### **2.Throughput**

Hashing and status look-ups are instant. In the test we cleared all 25 files in under a minute; the only wait time is the human click to approve or reject. A clerk needs 2–3 minutes per file to compare stamps, print logs, and sign off—roughly 20–30 documents an hour at best. Even on an ordinary laptop the new workflow can handle hundreds of documents an hour, freeing staff from repetitive visual checks.

#### **3. Scalability**

Uses plain SHA-256 and a simple “append-only” ledger. It runs fine on a single database today, and we can flip the ledger onto a permissioned blockchain later without rewriting code. The digital design can scale up city- or province-wide just by adding storage and a few web servers; no proportional jump in staff or storage rooms.

## **4. Data integrity**

Every file's hash is locked in the ledger the moment it is issued. If a byte changes, the hash changes, and the system turns the status red. Users always pull the unedited original from protected storage. The SHA verification approach gives us tamper-evidence by default: a permanent, verifiable record that proves each document is the same today as the day it left the office.

# **CHAPTER 6**

## **CONCLUSION**

### **6.1 Research Findings and Conclusion**

This project aimed to develop a Secure E-Documentation Verification System that allows users to upload important documents and get them verified by authorized personnel using cryptographic techniques such as SHA-256 hashing and QR code validation. After conducting a thorough analysis and implementation, the system successfully met its core objectives, Ensuring document integrity by generating unique hash values at the time of upload. Preventing tampering and duplication through hash comparison and QR-based verification. Providing a user-friendly dashboard for admins to track the status of documents (e.g., verified, pending, duplicate, rejected, tampered). Enhancing transparency and trust in digital document validation workflows.

The system's ability to detect tampered and duplicate documents demonstrates the reliability of the verification process. Moreover, the real-time feedback and status categorization streamline the interaction between users and verifiers. The project contributes to creating a more secure and efficient ecosystem for digital document validation, especially in contexts such as academic certificates, identity documents, and legal papers.

### **6.2 Scope for Further Enhancement:**

1. Put the records on a blockchain. Linking each document's hash to a private blockchain would lock the history in place so nothing can ever be changed behind the scenes.
2. Add one-time codes or email checks. Sending OTP on a phone or confirmation link to the inbox provides an extra layer of protection when someone uploads or approves a file
3. Let the system read the pages for us. Using OCR and basic AI, the software could spot blank spots, missing stamps, or other oddities before a human even looks at the document.

4. Move storage to the cloud. Hosting the database on Firebase, AWS, or a similar service means the app can grow from dozens to thousands of users without slowing down.
5. Build a phone app. A lightweight mobile version would let people snap a photo of a paper form, upload it, and get a quick “Verified” status from anywhere.
6. Put the records on a shared chain. Linking each document’s hash to a private blockchain would lock the history in place so nothing can ever be changed behind the scenes.
7. Add one-time codes or email checks. Sending an OTP to a phone or a confirmation link to an inbox gives an extra layer of security whenever someone uploads or approves a file.
8. Move storage to the cloud. Hosting a database on Firebase, AWS or similar service means the app can grow from dozens to thousands of users, without slowing down.

# APPENDICES

## APPENDIX-1:

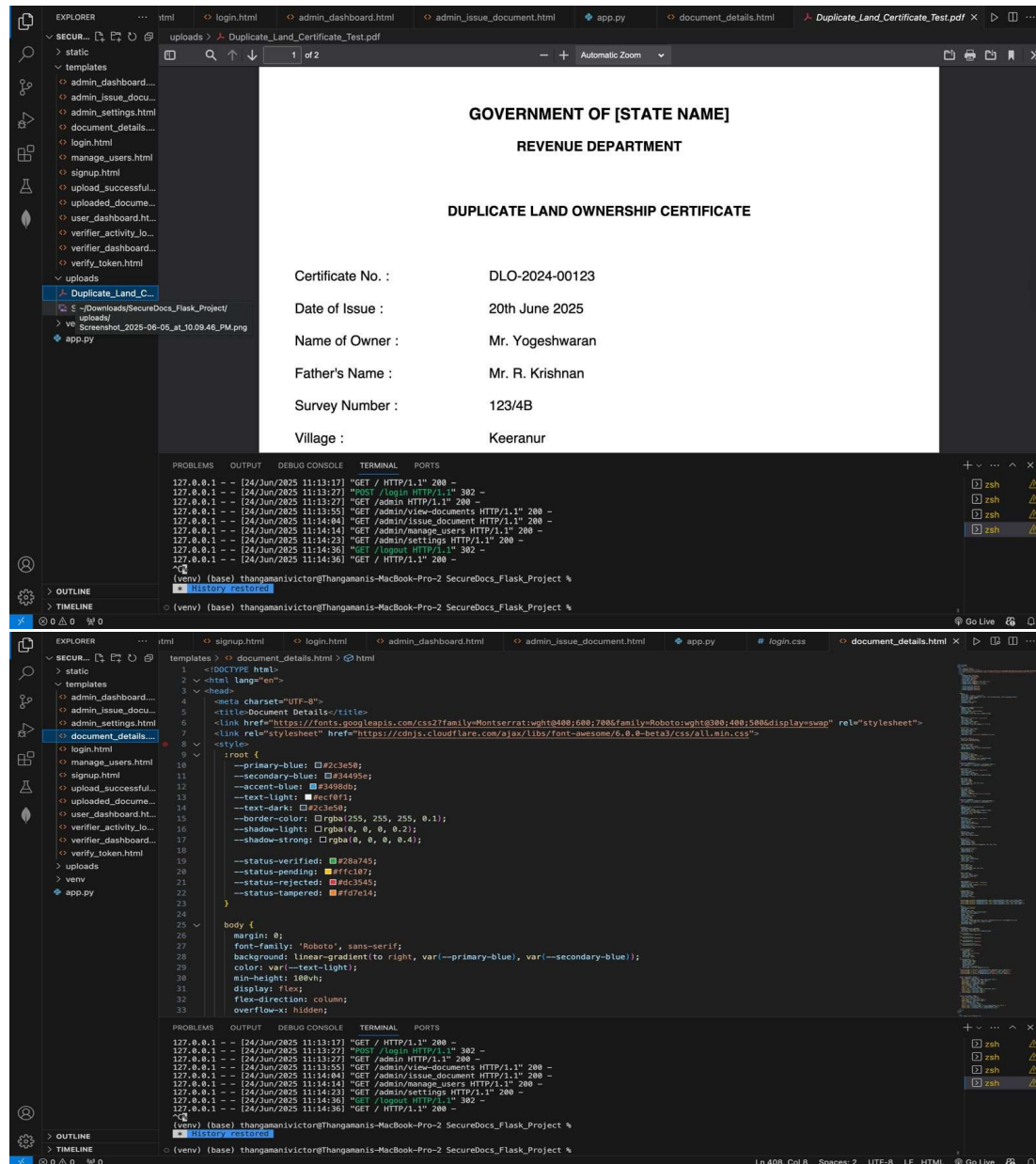
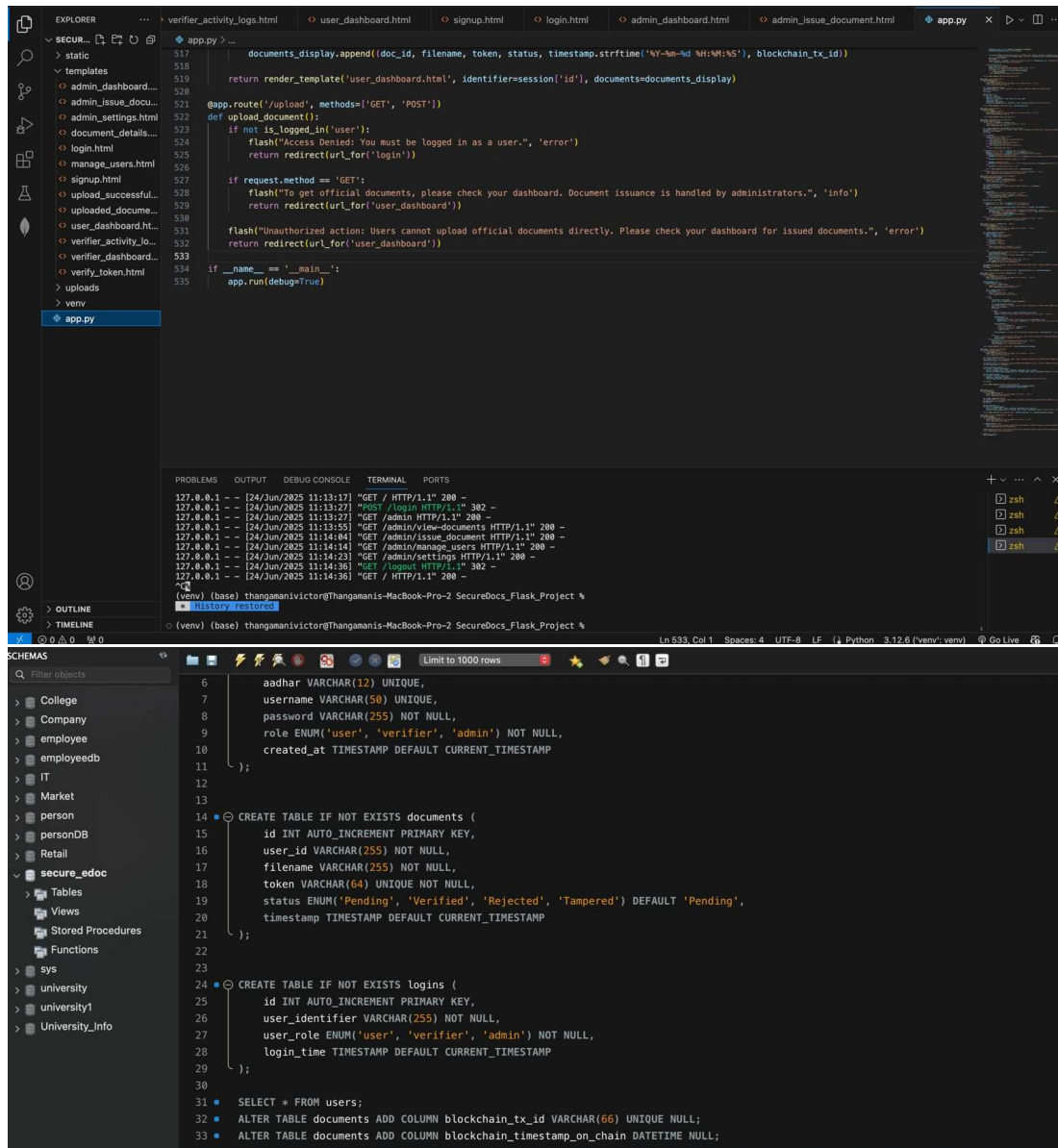
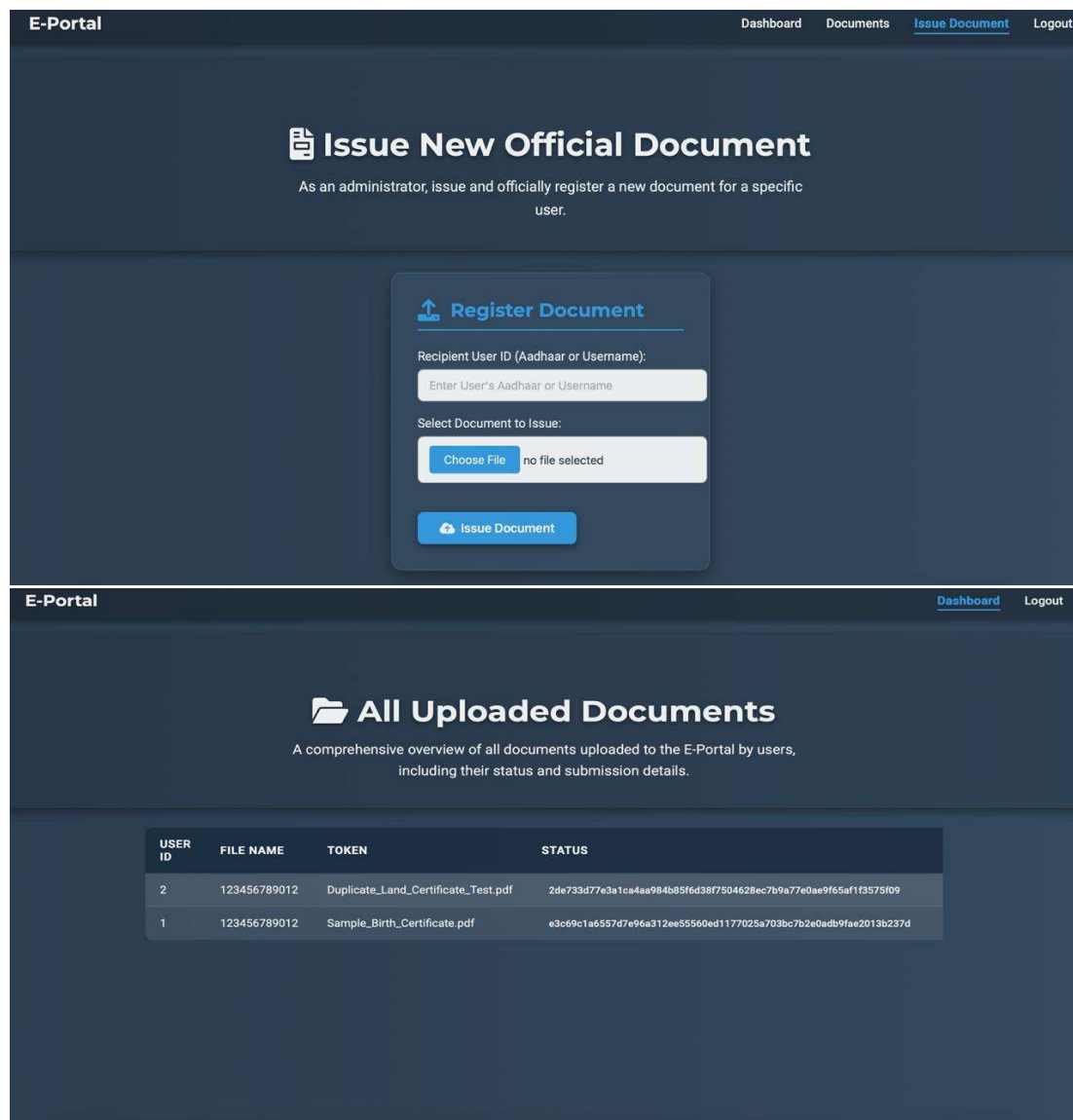


Fig: A1 Uploads



**Fig A2 Sql configuration**



**Fig A3&A4 Admin dashboard**

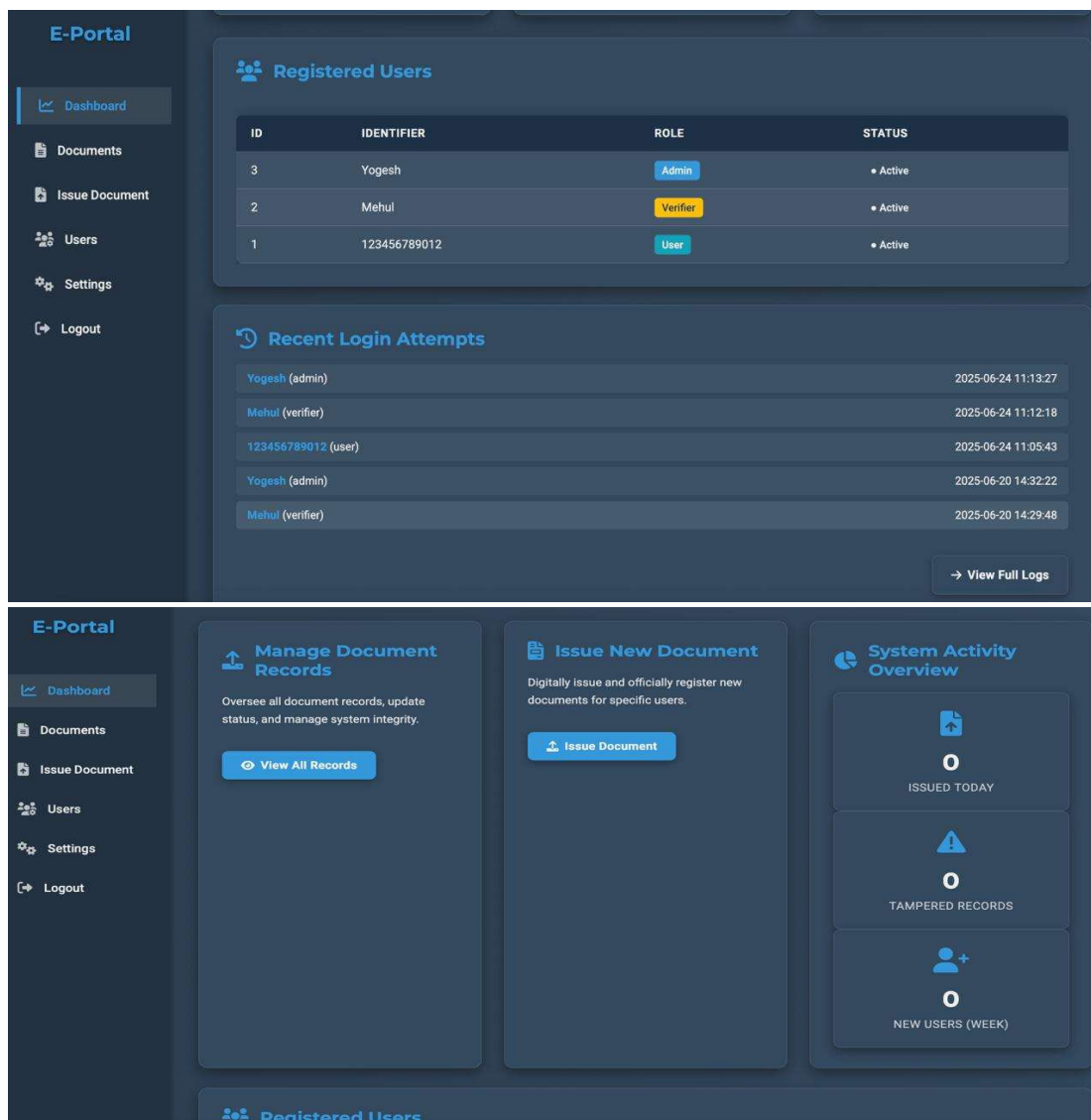
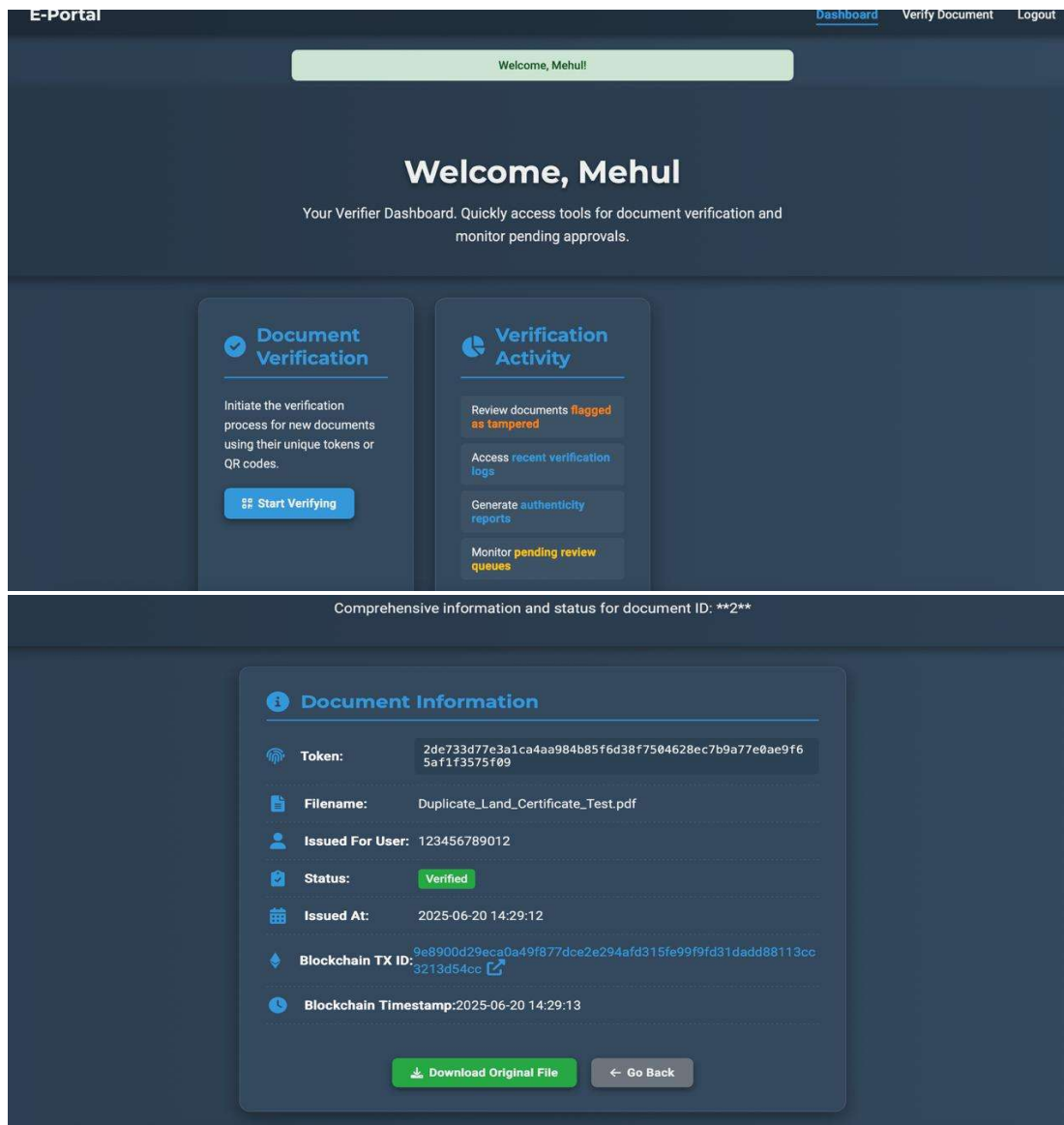
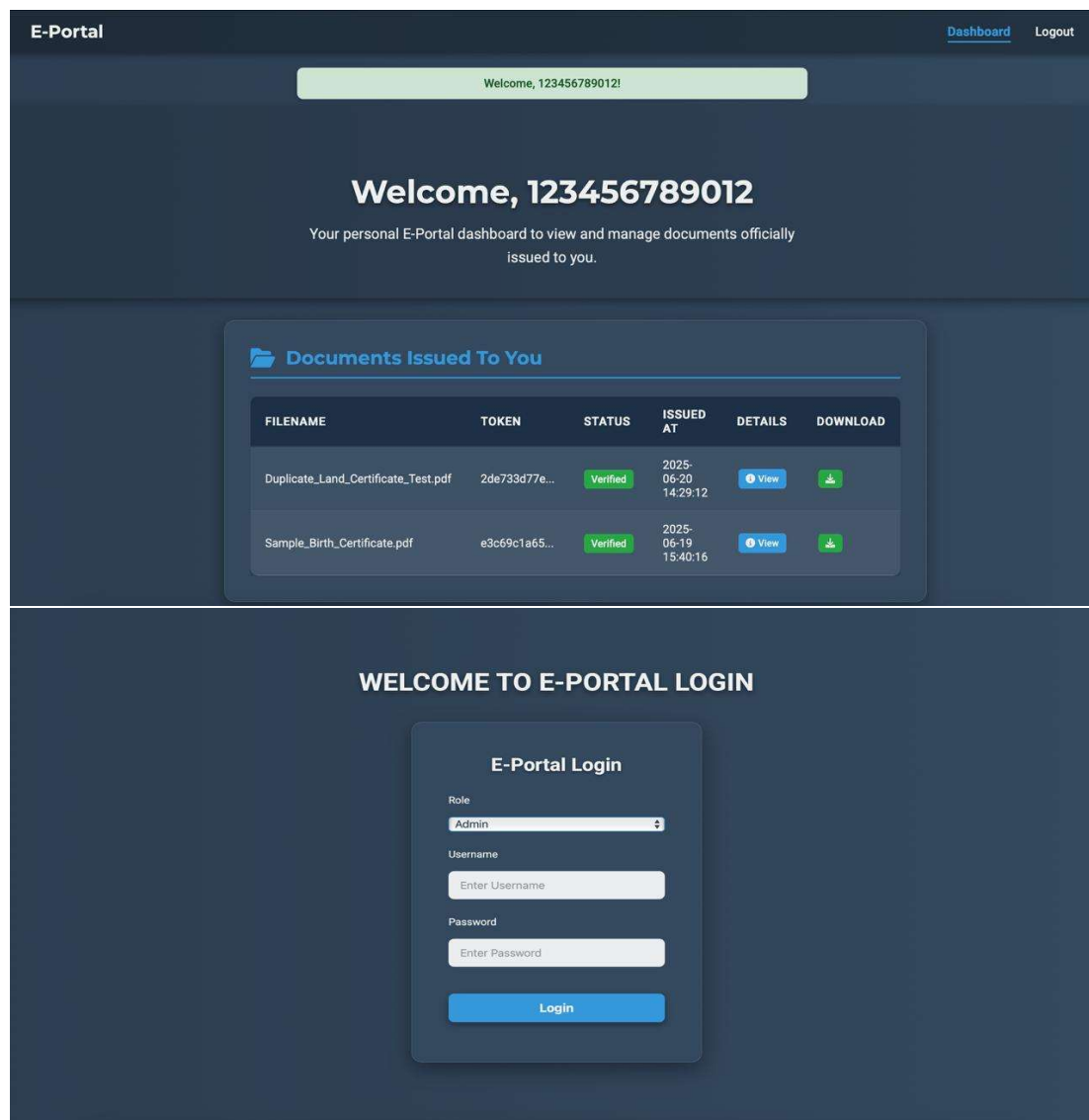


Fig A5 Verifier Dashboard



**Fig A6 Admin Dashboard**



**Fig A7 User Dashboard and Login Page**